RoomSage: An AI-Powered Interior Redesign and 3D Room Planning Framework

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Abstract

Architectural interior redesign and automated room planning traditionally suffer from a dual bottleneck: a visualization gap, where generic generative AI image tools hallucinate physically impossible layouts and non-existent geometry, and an execution gap, where aesthetic renders cannot be translated into actionable, budget-bounded procurement lists in local currencies. This paper presents RoomSage, a proposed five-stage hybrid spatial framework that unifies classical Structure-from-Motion (COLMAP) and 3D Gaussian Splatting with spatially conditioned generative diffusion (SDXL + ControlNet), text-guided 3D scene editing (Instruct-GS2GS), single-view 3D asset generation (TripoSR), and an LLM-driven budget transformation planner. RoomSage is designed to ingest 5–15 overlapping mobile camera photographs, reconstruct an explorable "Before" 3D Gaussian Splatting model, enable rapid 2D style exploration via depth-conditioned ControlNet diffusion, and support direct 3D scene editing with catalog-matched furniture placement under explicit budget constraints. The framework is intended to bridge generative interior concepts and physical architectural execution, with emphasis on spatial fidelity and consumer-GPU accessibility.

Index Terms— 3D Gaussian Splatting, ControlNet Diffusion, Interior Design Automation, Structure-from-Motion, Spatial AI, LLM Budget Optimization, Mobile Computer Vision.

I. INTRODUCTION AND PROBLEM STATEMENT

Residential interior renovation remains constrained by modern visual tools. Homeowners, interior architects, and real-estate stakeholders frequently encounter two persistent gaps: the visual hallucination gap and the financial execution gap.

First, state-of-the-art 2D text-to-image generative models excel at producing aesthetically striking concepts. However, when prompted with room redesign requests, unconstrained diffusion models tend to distort wall dimensions, fabricate door and window placements, alter camera perspective across successive generations, and disregard existing room geometry. Consequently, generic 2D generative output cannot be trusted as a valid representation of a user's actual physical room.

Second, traditional 3D CAD and manual modeling tools require specialized training, multi-hour alignment, and extensive asset curation. Even when a realistic 3D render is produced, existing platforms typically lack an automated bridge linking visual furniture models to real-world e-commerce catalogs, regional currency pricing, and hard budgetary caps specified by the user.

RoomSage is proposed as a unified five-stage hybrid spatial computing framework intended to address both gaps. The system is designed so that users capture 5–15 casual smartphone photographs of their room; the pipeline then constructs a 3D Gaussian Splatting model, generates spatially constrained 2D redesign previews, supports direct 3D scene editing and object replacement, and produces an itemized procurement plan grounded in a real product catalog.

Core research contributions of the proposed framework include:

- **Spatial Fidelity Guarantee:** Generative diffusion is grounded on 3D Gaussian Splatting geometry and depth maps rendered from the user's own camera capture.
- **End-to-End Actionability:** Redesigned items are cross-referenced against a structured vendor product catalog with local-currency constraints and purchase links.
- **Consumer GPU Accessibility:** The pipeline is engineered with model CPU offloading, attention slicing, and half-precision execution so that it can run within a modest VRAM envelope (targeting entry-level consumer GPUs).

II. RELATED WORK AND THEORETICAL FOUNDATIONS

A. 3D Gaussian Splatting and Indoor Scene Reconstruction

Radiance field representations have evolved from implicit Neural Radiance Fields (NeRF) to explicit 3D Gaussian Splatting (3DGS). Kerbl et al. [1] introduced 3DGS, representing scenes with anisotropic 3D Gaussians characterized by position, covariance, opacity, and spherical harmonics. While 3DGS achieves real-time rasterization and high rendering fidelity, indoor reconstruction remains challenging due to textureless walls, wide camera motions, and unconstrained lighting. Extensions such as FreeSplat [4] and GaussianRoom [3] explore cross-view aggregation and SDF-guided constraints to reduce floaters in large-scale indoor captures. RoomSage leverages COLMAP structure-from-motion [10] together with optimized 3DGS implementations (gsplat / splatfacto) to establish structurally grounded room primitives.

B. Spatially Conditioned Generative Diffusion

ControlNet [2] locks pretrained UNet weights while training encoding copies connected via zero convolutions, enabling structural conditioning with edge maps, poses, and monocular depth. In RoomSage, depth-conditioned ControlNet coupled with Stable Diffusion XL (SDXL) is used to lock generative interior renders to the 3D depth geometry extracted from the user's real room reconstruction, reducing spatial hallucination.

C. Text-Driven 3D Scene Editing and Asset Synthesis

Modifying explicit 3D Gaussian scenes without full retraining is an active research direction. Instruct-GS2GS [5] combines 2D image-to-image editing guidance with iterative 3DGS optimization to maintain multi-view consistency for style and material changes. For discrete object insertion, feed-forward single-view 3D

generators such as TripoSR [8] enable rapid mesh reconstruction from catalog product images, supporting compositional placement into the splat scene. Related systems include GaussCtrl [6] and GaussianEditor [7].

III. METHODOLOGY AND SYSTEM ARCHITECTURAL PIPELINE

The RoomSage framework is structured as a five-stage sequential and iterative hybrid pipeline. Each stage performs specialized computer-vision or language tasks and transfers structured artifacts via an asynchronous job queue.

Stage	Core Tech	Output
1. Ingestion & QA	OpenCV, VLM	Normalized JPEGs, Scene JSON
2. “Before” 3DGS	COLMAP, gsplat, SAM	before_model.splat, depth map, masks
3. 2D Redesign	SDXL + ControlNet	2D Redesign Preview PNG
4. “After” 3D Edit	Instruct-GS2GS, TripoSR	after_model.splat, composite scene
5. LLM Planner	Claude / GPT-4, Catalog DB	Shopping list, costs, URLs

Table I: Overview of the five-stage RoomSage pipeline (planned architecture).

A. Stage 1: Mobile Photo Ingestion, Quality Assurance, and Scene Understanding

Users begin with guided multi-photo capture (5–15 overlapping photographs) through a Progressive Web App (PWA). Uploaded images are passed through an automated Quality Assurance service that performs: (1) EXIF orientation correction and resolution capping at 4K to control VRAM during feature extraction; (2) Laplacian variance blur detection, flagging images below a sharpness threshold; (3) exposure analysis based on mean pixel intensity; and (4) vision-language scene captioning that produces a structured JSON summary of room type, existing furniture, layout, and lighting conditions. This JSON is reused by every later stage.

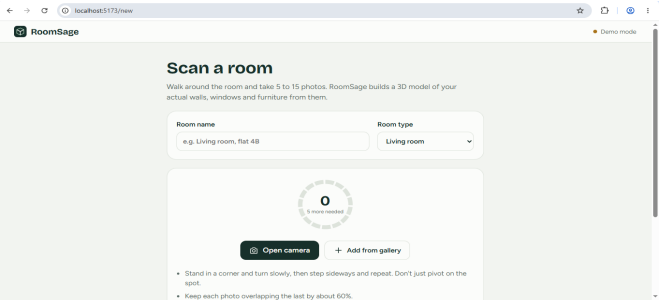


Fig. 1: Prototype capture interface — room naming, type selection, and guided multi-photo instructions.

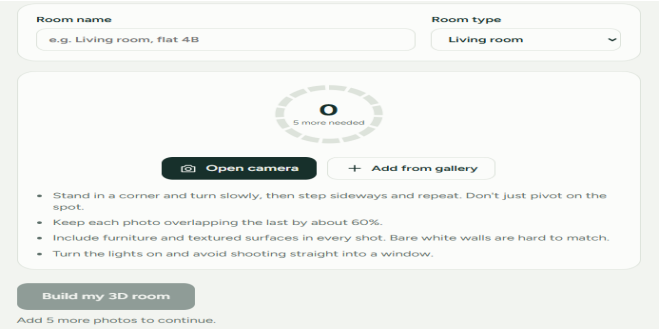


Fig. 2: Capture guidance checklist and “Build my 3D room” action.

B. Stage 2: “Before” 3D Scene Reconstruction and Geometry Extraction

Camera poses and a sparse point cloud are computed with COLMAP structure-from-motion across the validated photo set. A 3D Gaussian Splatting model is then optimized (gsplat / splatfacto). The trained model renders a normalized 8-bit depth map that serves as the geometric anchor for subsequent stages. Primary furniture instances are segmented with SAM [9] to enable object-level editing later.

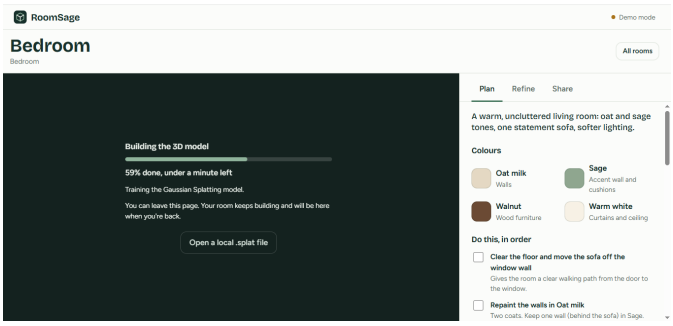


Fig. 3: Prototype project view while the “Before” 3D Gaussian Splatting model is being built, with progress feedback and the parallel plan panel.

C. Stage 3: Rapid 2D Redesign Preview and Chat-Based Style Iteration

To keep early exploration responsive, Stage 3 uses 2D generative diffusion. The depth map from Stage 2 conditions a ControlNet + SDXL pipeline. Users supply natural-language design prompts (for example, “warm Scandinavian study under a fixed budget”). Because generation is strictly conditioned on the real room’s depth geometry, the resulting previews remain spatially aligned. Users can iteratively refine colors, materials, and themes through short conversational turns before committing to full 3D editing.

D. Stage 4: “After” 3D Scene Editing and Catalog Asset Composition

Once a 2D preview is approved, Stage 4 applies style and material changes directly to the original 3D Gaussian model via Instruct-GS2GS (an iterative render–edit–optimize loop). For discrete furniture replacement, segmented objects are removed and catalog items are converted into 3D meshes with TripoSR, then placed at target coordinates consistent with product dimensions. This compositional approach reuses the same underlying reconstruction rather than regenerating independent 2D views and re-fusing them.

E. Stage 5: LLM Transformation Planner and Catalog Budget Engine

Stage 5 converts visual designs into a purchasing action plan. An LLM planner consumes the scene JSON, user budget cap, and design theme, then queries a structured product catalog that stores real titles, categories, local-currency prices, dimensions, vendor URLs, and isolated product images. Wherever a catalog match exists, the shopping list uses actual purchasable prices; otherwise estimated ranges are clearly distinguished. When a budget constraint is supplied, the system checks the running total and can suggest cheaper alternatives or items to drop, rather than asserting an unverified total.

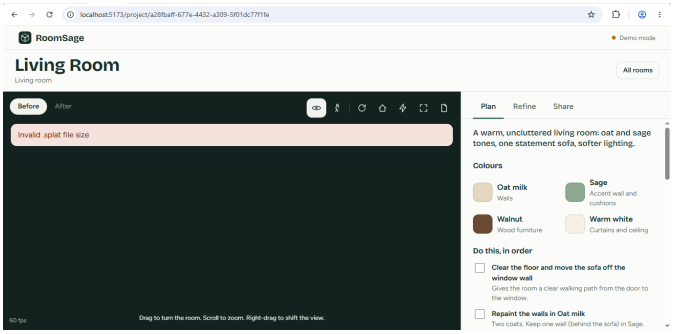


Fig. 4: Prototype project workspace showing the 3D viewer area (Before / After), plan panel with color palette, prioritized action steps, and refinement / share tabs.

IV. SYSTEMS INFRASTRUCTURE AND DESIGN CONSIDERATIONS

RoomSage is designed around a decoupled, asynchronous microservice backend so that long-running 3D reconstruction and editing jobs do not block the interactive front end.

- **Backend:** FastAPI orchestrates the pipeline; Celery workers backed by Redis handle asynchronous tasks; PostgreSQL stores users, projects, and refinement history; S3-compatible object storage holds photos, splats, depth maps, and generated assets.
- **VRAM-conscious execution:** Heavy diffusion and 3DGS stages are intended to run with model CPU offloading, attention slicing, and fp16 precision so that the pipeline remains usable on consumer GPUs with limited VRAM.
- **State and workflow guards:** Depth-map availability is required before 2D redesign; 3D editing is gated on an explicit preview-approval record referencing a non-stale 2D preview and valid catalog product keys.
- **Frontend:** A Progressive Web App (React + Vite or Next.js) provides guided multi-photo capture via browser camera APIs, a Three.js / React Three Fiber viewer for walkable "Before" and "After" models, status polling for long jobs, and a natural-language refinement interface.

V. FEATURES AND DELIVERABLES

The planned core features of RoomSage are:

- Multi-photo room capture with in-app guidance for sufficient 3D reconstruction coverage.
- Explorable "Before" 3D model reconstructed from the user's actual room geometry.
- Text-to-redesign via natural-language style, budget, and functional prompts.
- Spatially consistent "After" 3D model obtained by editing the reconstructed scene.
- Fast 2D before/after shareable preview for social and collaborative use.
- Structured transformation plan: prioritized rearrangement steps, color palette, and a shopping list with local-currency cost estimates and purchase links where catalog matches exist.
- Iterative natural-language refinement that updates both the visual design and the plan without requiring a full re-capture.

Differentiating aspects of the design include object-level furniture swap grounded in real purchasable products, budget-constrained planning that re-optimizes against an explicit cap, and true spatial fidelity obtained by conditioning generation on the user's own room geometry.

Stretch goals beyond the initial implementation include AR preview of the redesigned room in the physical space, multi-room projects with shared style continuity, and collaborative project sharing.

VI. SUGGESTED IMPLEMENTATION ORDER

Given a constrained timeline and shared GPU resources, the build order prioritizes a reliable reconstruction baseline first, then generative and planning layers:

1. Multi-photo capture UX + COLMAP + splatfacto reconstruction → reliable "Before" model (serves as the primary demo even if later stages are incomplete).
2. SAM segmentation + TripoSR object generation and placement.
3. ControlNet + SDXL 2D redesign + LLM planner (largely API and prompt engineering).
4. Instruct-GS2GS integration for style-based "After" 3D edits.
5. Refinement loop, PWA polish, and 3D viewer UX.

Stages that rely on generative models (scene captioning, 2D redesign, Instruct-GS2GS style editing, TripoSR object generation, and the LLM planner) are clearly separated from classical computer-vision stages (COLMAP pose estimation, photometric 3DGS fitting, and discriminative SAM segmentation). This separation allows parallel progress once the reconstruction baseline is stable.

VII. LIMITATIONS AND FUTURE DIRECTIONS

Several practical constraints are anticipated. Specular highlights or dynamic shadows during capture can introduce Gaussian floaters near reflective surfaces. Highly featureless walls may produce localized COLMAP sparsity; monocular normal or depth priors can help mitigate this. Full WebXR AR walkthroughs would require progressive level-of-detail splat streaming over mobile networks. Future work may explore feed-forward generalizable Gaussian Splatting backbones to reduce Stage-2 training times, tighter integration of localized editors (e.g., GaussianEditor-style single-object recoloring), and multi-room style consistency.

VIII. CONCLUSION

RoomSage is proposed as a framework that bridges inspirational AI room generation and real-world physical renovation. By grounding generative diffusion on 3D Gaussian Splatting geometry and depth maps derived from the user's own photographs, and by connecting visual edits to a vendor-backed product catalog with explicit budget enforcement, the system aims to deliver spatially accurate, commercially actionable interior redesign guidance suitable for consumer mobile devices and modest GPU hardware. The architecture, capture guidance, viewer, and planning surfaces described above form the concrete implementation plan that the project will follow.

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